

Polynomials

Expand, factorise, and read a cubic straight off its graph. Roots give brackets; one point gives the number in front.



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THE TOPIC ON ONE MAP

Polynomials

01 — Expand and factorise

three brackets

factorise completely

02 — Solving cubics

show all stages

hidden cubics

$f(-1) = 32$

03 — Cubic from a graph

cuts / touches

$f(x - 3) = 0$

01 Expand and Factorise

Expand: every term $\times$ every term. Factorise: HCF first, then the CC Rule on what is left.

WHEN YOU SEE $\rightarrow$ YOU DO

“expand and simplify” three brackets

$\rightarrow$ two first, then the third

“factorise completely” a cubic with no constant

$\rightarrow$ take out x , then factorise the quadratic

“(3x + y)(2x - 3y)(x - 2y) $\equiv$ ax³ + bx²y + ...”

$\rightarrow$ expand, compare coefficients

x⁵y - xy⁵

$\rightarrow$ HCF, then DOTS twice

WORKED MODEL · FACTORISE COMPLETELY

$$3x^3 - 51x^2 + 126x$$

1 $3x(x^2 - 17x + 42)$

HCF 3x

2 roots of the quadratic: 3, 14

calculator

ANSWER $3x(x - 3)(x - 14)$

IDEA 01

Practise: Expand and Factorise

My questions first, then selected practice.

Q1 **DR ESLAM** HCF, then CC Rule

P1 · TOPIC 5 2 marks

Factorise completely (a) $x^3 - 4x$ (b) $2x^3 + x^2 - 6x$

SELECTED PRACTICE**Q2** **PRACTICE** Practice · Easy

P1 · TOPIC 5 5 marks

2 Expand and simplify

(i) $(2x + 3)(x - 4)$

(ii) $2p(p + 3)(p - 2)$

(iii) $(y - 1)(y - 2)^2$

(5 marks)

Q3**PRACTICE**

Practice · Hard

P1 · TOPIC 5**2 marks**

5 Factorise completely $15x^3 + 19x^2 - 10x$.

(2 marks)**Q4****PRACTICE**

Practice · Very Hard

P1 · TOPIC 5**3 marks**

5 Factorise completely $x^5y - xy^5$.

(3 marks)

02 Solving Cubics and Hidden Cubics

Factorise to brackets, each bracket = 0. A power pattern like y , $y^{\frac{2}{3}}$, $y^{\frac{1}{3}}$ is the same cubic in disguise.

WHEN YOU SEE → YOU DO

“solve $f(x) = 0$ ” (show all stages)

→ factorise fully, each bracket = 0

“hence solve $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$ ”

→ $x = y^{\frac{1}{3}}$ — the same cubic

“ $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0$ ”

→ $x = 9^z$; reject $x \leq 0$

“given $f(-1) = 32$, show $a = 3$ ”

→ substitute, solve for a

WORKED MODEL · OCTOBER 2022

$$f(x) = x^2(2x + 1) - 15x$$

1 $x(2x^2 + x - 15) = x(2x - 5)(x + 3) = 0$

factorise

2 $x = 0, \frac{5}{2}, -3$

roots

3 hence $y^{\frac{4}{3}}(2y^{\frac{2}{3}} + 1) - 15y^{\frac{2}{3}} = 0$: $x = y^{\frac{2}{3}} = \frac{5}{2}$

$y > 0$

ANSWER $y = \left(\frac{5}{2}\right)^{\frac{3}{2}} = \frac{5\sqrt{10}}{4}$

IDEA 02

Practise: Solving Cubics

My questions first, then real past papers.

Q5 **DR ESLAM** Solve a cubic

P1 · TOPIC 5 3 marks

Given that $x = 1$ is one root, solve $x^3 - 7x + 6 = 0$ by factorising.

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · June 2021 · Q2

P1 · TOPIC 5 10 marks

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

where a is a positive constant.

Given $f(-1) = 32$

(a) (i) show that the only possible value for a is 3

(ii) Using $a = 3$ solve the equation

$$f(x) = 0 \quad (5)$$

(b) Hence find all real solutions of

(i) $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$

(ii) $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0 \quad (5)$

(Total 10 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q4

P1 · TOPIC 5**6 marks**

4.

**In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.**

$$f(x) = x^2(2x + 1) - 15x$$

(a) Solve

$$f(x) = 0$$

(4)

(b) Hence solve

$$y^{\frac{4}{3}}\left(2y^{\frac{2}{3}} + 1\right) - 15y^{\frac{2}{3}} = 0 \quad y > 0$$

giving your answer in simplified surd form.

(2)**(Total for Question 4 is 6 marks)**

03 Cubic from a Graph

Crossing the x -axis at $a \rightarrow (x - a)$. **Touching** at $a \rightarrow (x - a)^2$. Then one more point finds the number in front, k .

WHEN YOU SEE $\rightarrow$ YOU DO

“cuts the x -axis at $-4, -2, 3$ ”

$\rightarrow y = k(x + 4)(x + 2)(x - 3)$

“touches the x -axis at 5 ”

$\rightarrow$ repeated root $(x - 5)^2$

“cuts the y -axis at 10 ” / “passes through P ”

$\rightarrow$ substitute to find k

“roots of $f(\frac{1}{3}x) = 0$ ”, “ $f(x - 3) = 0$ ”

$\rightarrow$ stretch $\times 3$ · shift right 3

curve meets a line

$\rightarrow$ curve = line; you already know one root $\rightarrow$ divide it out

WORKED MODEL · JANUARY 2026

Cuts at -2 , touches at 5 , y -intercept 10 .

1 $f(x) = k(x + 2)(x - 5)^2$

roots $\rightarrow$ brackets

2 $10 = k(2)(25) \Rightarrow k = \frac{1}{5}$

y -intercept

3 $\frac{1}{5}(x + 2)(x - 5)^2 = 10(x + 2) \Rightarrow (x - 5)^2 = 50$

meets $y = 10(x + 2)$

ANSWER $x = -2, \quad x = 5 \pm 5\sqrt{2}$

IDEA 03

Practise: Cubic from a Graph

My questions first, then real past papers.

Q8 DR ESLAM Roots to equation

P1 · TOPIC 5 3 marks

A cubic cuts the x-axis at -1 , 2 and 3 and passes through $(0, 6)$. Find its equation.

Q9 DR ESLAM Touching root

P1 · TOPIC 5 3 marks

A cubic touches the x-axis at $x = 2$, cuts it at $x = -1$ and passes through $(0, 8)$. Find its equation in factorised form.

PAST PAPERS

Q10 PAST PAPER Pearson IAL WMA11 · June 2021 · Q8

P1 · TOPIC 5 9 marks

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

- (a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point P is the minimum point of C_1

- (b) Deduce the coordinates of P .

(1)

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A , B , C and D are constants.

Given that C_2

- passes through P
- intersects the x -axis at -4 , -2 and 3

- (c) find, making your method clear, the values of A , B , C and D .

(5)

(Total 9 marks)

4.

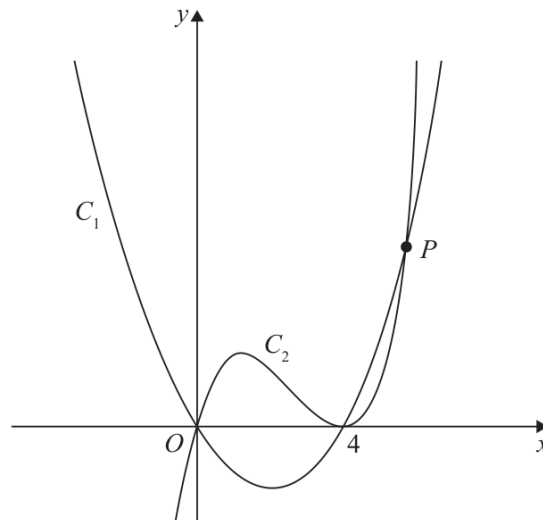


Figure 1

Figure 1 shows a sketch of part of the curves C_1 and C_2

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a quadratic function
- cuts the x -axis at the origin and at $x = 4$
- has a minimum turning point at $(2, -4.8)$

(a) find $f(x)$

(3)

Given that C_2

- has equation $y = g(x)$ where $g(x)$ is a cubic function
- cuts the x -axis at the origin and meets the x -axis at $x = 4$
- passes through the point $(6, 7.2)$

(b) find $g(x)$

(3)

The curves C_1 and C_2 meet in the first quadrant at the point P , shown in Figure 1.

(c) Use algebra to find the coordinates of P .

(4)

(Total for Question 4 is 10 marks)

Q12 PAST PAPER Pearson IAL WMA11 · January 2026 · Q6

P1 · TOPIC 5 9 marks

6.

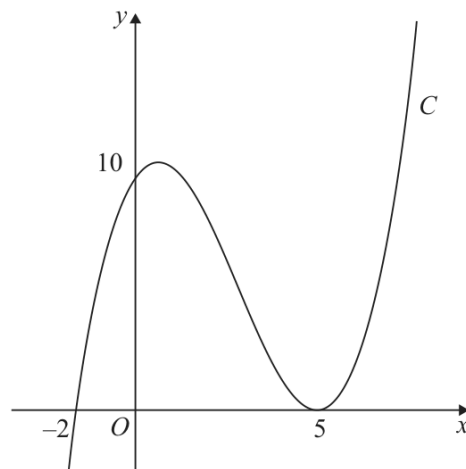
**Figure 3**

Figure 3 shows a sketch of the curve C with equation $y = f(x)$ where $f(x)$ is a cubic function in x .

The curve C

- cuts the x -axis at $(-2, 0)$ and cuts the y -axis at $(0, 10)$
- touches the x -axis at $(5, 0)$

as shown in Figure 3.

(a) Deduce the roots of the equation

(i) $f\left(\frac{1}{3}x\right) = 0$

(ii) $f(x - 3) = 0$

(2)

(b) Find an expression for $f(x)$. You should leave your answer in factorised form.

(3)

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(c) Use algebra to find the exact x coordinates of the three points of intersection.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 9 marks)

Q13 PAST PAPER Pearson IAL WMA11 · June 2026 · Q6

P1 · TOPIC 5 7 marks

6.

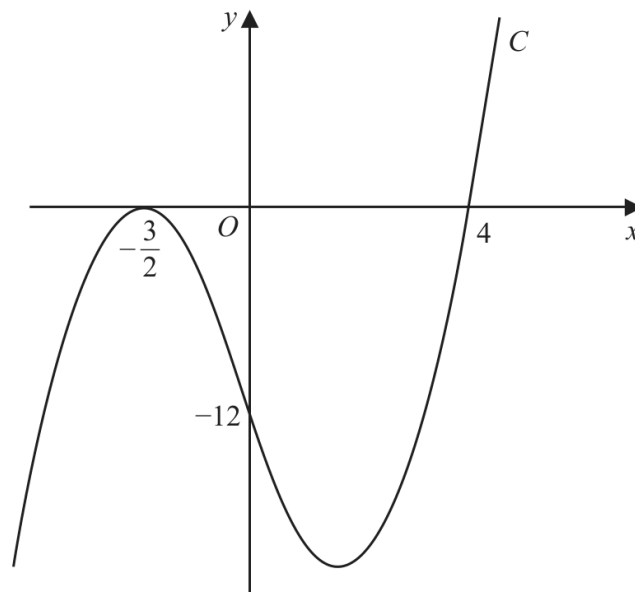
**Figure 2**

Figure 2 shows a sketch of the curve C with equation $y = f(x)$.

The curve C

- touches the x -axis at $-\frac{3}{2}$
- cuts the x -axis at 4
- cuts the y -axis at -12

Given that $f(x)$ is a cubic function,

(a) find $f(x)$. You may leave the answer in a factorised form.

(3)

The line with equation $y = 2x - 12$ cuts C at $(0, -12)$ and at two further points, P and Q .

(b) Use algebra to find the exact x coordinates of P and Q .

Give each answer in simplest form.

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

Question 6 continued

(Total for Question 6 is 7 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Expand / fac-
torise** Every term $\times$ every term. HCF first, then CC Rule.
Compare coefficients.
- 02 Solve** Factorise fully, each bracket = 0. Hidden cubic:
 $x = y^{\frac{1}{3}}$, $x = 9^z$.
- 03 From a graph** Cut $\rightarrow (x - a)$, touch $\rightarrow (x - a)^2$, a point $\rightarrow k$.

BOTTOM LINE Roots $\rightarrow$ brackets $\rightarrow$ one point for k . Touching means squared.

Answers

Polynomials • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • EXPAND AND FACTORISE

- Q1** Dr Eslam • HCF, then CC Rule (a) $x(x+2)(x-2)$ (b) $x(2x-3)(x+2)$
- Q2** Practice • Easy (i) $2x^2 - 5x - 12$ (ii) $2p^3 + 2p^2 - 12p$ (iii) $y^3 - 5y^2 + 8y - 4$
- Q3** Practice • Hard $x(3x+5)(5x-2)$
- Q4** Practice • Very Hard $xy(x-y)(x+y)(x^2+y^2)$

IDEA 02 • SOLVING CUBICS

- Q5** Dr Eslam • Solve a cubic $(x-1)(x-2)(x+3): x = 1, 2, -3$
- Q6** Pearson IAL WMA11 • June 2021 • Q2 (a)(ii) $x = 0, \frac{1}{3}, -9$ (b)(i) $y = 0, \frac{1}{27}, -729$ (ii) $z = -\frac{1}{2}$
- Q7** Pearson IAL WMA11 • October 2022 • Q4 (a) $x = 0, \frac{5}{2}, -3$ (b) $y = \frac{5\sqrt{10}}{4}$

IDEA 03 • CUBIC FROM A GRAPH

- Q8** Dr Eslam · Roots to equation $y = (x + 1)(x - 2)(x - 3) = x^3 - 4x^2 + x + 6$
- Q9** Dr Eslam · Touching root $y = 2(x + 1)(x - 2)^2$
- Q10** Pearson IAL WMA11 · June 2021 · Q8 (a) $3(x + 1)^2 + 6$ (b) $P(-1, 6)$ (c) $A = -\frac{1}{2}$, $B = -\frac{3}{2}$, $C = 5$, $D = 12$
- Q11** Pearson IAL WMA11 · October 2024 · Q4 (a) $f(x) = \frac{6}{5}x^2 - \frac{24}{5}x$ (b) $g(x) = \frac{3}{10}x(x - 4)^2$ (c) $P(8, 38.4)$
- Q12** Pearson IAL WMA11 · January 2026 · Q6 (a)(i) $x = -6, 15$ (ii) $x = 1, 8$ (b) $f(x) = \frac{1}{5}(x + 2)(x - 5)^2$ (c) $x = -2, 5 \pm 5\sqrt{2}$
- Q13** Pearson IAL WMA11 · June 2026 · Q6 (a) $f(x) = \frac{1}{3}(2x + 3)^2(x - 4)$ (b) $x = \frac{1 \pm \sqrt{46}}{2}$